

Java Banking System Using OOP

```
import java.util.ArrayList;
import java.util.List;

class Account {
    private String accountNumber;
    private String ownerName;
    private double balance;

    public Account() {
        this("0000", "Unknown", 0.0);
    }

    public Account(String accountNumber, String ownerName, double balance) {
        this.accountNumber = accountNumber;
        this.ownerName = ownerName;
        this.balance = balance;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public void setAccountNumber(String accountNumber) {
        this.accountNumber = accountNumber;
    }

    public String getOwnerName() {
        return ownerName;
    }

    public void setOwnerName(String ownerName) {
        this.ownerName = ownerName;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        if (balance < 0) {
            throw new IllegalArgumentException("Balance cannot be negative");
        }
        this.balance = balance;
    }

    public void deposit(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid deposit amount");
        }
        balance += amount;
    }

    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid withdrawal amount");
        }

        if (amount > balance) {
            throw new IllegalArgumentException("Insufficient balance");
        }

        balance -= amount;
    }

    public void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Owner Name: " + ownerName);
        System.out.println("Balance: " + balance);
    }
}

class SavingsAccount extends Account {
    private double interestRate;
}
```

```

    public SavingsAccount(String accountNumber,
                           String ownerName,
                           double balance,
                           double interestRate) {
        super(accountNumber, ownerName, balance);
        this.interestRate = interestRate;
    }

    public double calculateInterest() {
        return getBalance() * interestRate / 100.0;
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Interest Rate: " + interestRate + "%");
        System.out.println("Interest Amount: " + calculateInterest());
    }
}

class CurrentAccount extends Account {
    private double overdraftLimit;

    public CurrentAccount(String accountNumber,
                           String ownerName,
                           double balance,
                           double overdraftLimit) {
        super(accountNumber, ownerName, balance);
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid withdrawal amount");
        }

        if (amount > getBalance() + overdraftLimit) {
            throw new IllegalArgumentException(
                "Overdraft limit exceeded");
        }

        setBalance(getBalance() - amount);
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Overdraft Limit: " + overdraftLimit);
    }
}

public class BankingSystem {
    public static void main(String[] args) {

        List<Account> accounts = new ArrayList<>();

        accounts.add(
            new SavingsAccount(
                "101",
                "Alice",
                10000,
                5.0)
        );

        accounts.add(
            new CurrentAccount(
                "102",
                "Bob",
                5000,
                2000)
        );

        for (Account account : accounts) {
            account.display();
            System.out.println("-----");
        }
    }
}

```

